存在截距

$$\theta(t) = S_0 e^{-kt} + \theta_r \tag{1}$$

$$\frac{d\theta}{dt} = -kS_0 e^{-kt} \tag{2}$$

$$\frac{d\theta}{dt} = -k(\theta(t) - \theta_r) = -k\theta(t) + k\theta_r \tag{3}$$

无截距

$$\theta(t) = S_0 e^{-kt} \tag{4}$$

$$\frac{\mathrm{d}\theta}{\mathrm{dt}} = -k\theta \Longrightarrow \frac{d\theta}{\theta} = -kdt \tag{5}$$

$$\begin{gathered} \ln\theta = -kt + C \\ \theta = e^C e^{-kt} = S_0 e^{-kt} \end{gathered} \tag{6}$$